

System Administration and Maintenance

Week 2 Portfolio Project
Enterprise Infrastructure Planning for
a Startup Company

Submitted by:

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Part 1 – Company Profile

Company Name:

- ABC Startup Solutions

Nature of Business:

- ABC Startup Solutions is a newly established software development company that provides software development, web and mobile application development, and tech support, to small and medium-sized businesses.

Company Vision:

- To become a trusted company, known in delivering fast, efficient, reliable, and innovative solutions to businesses.

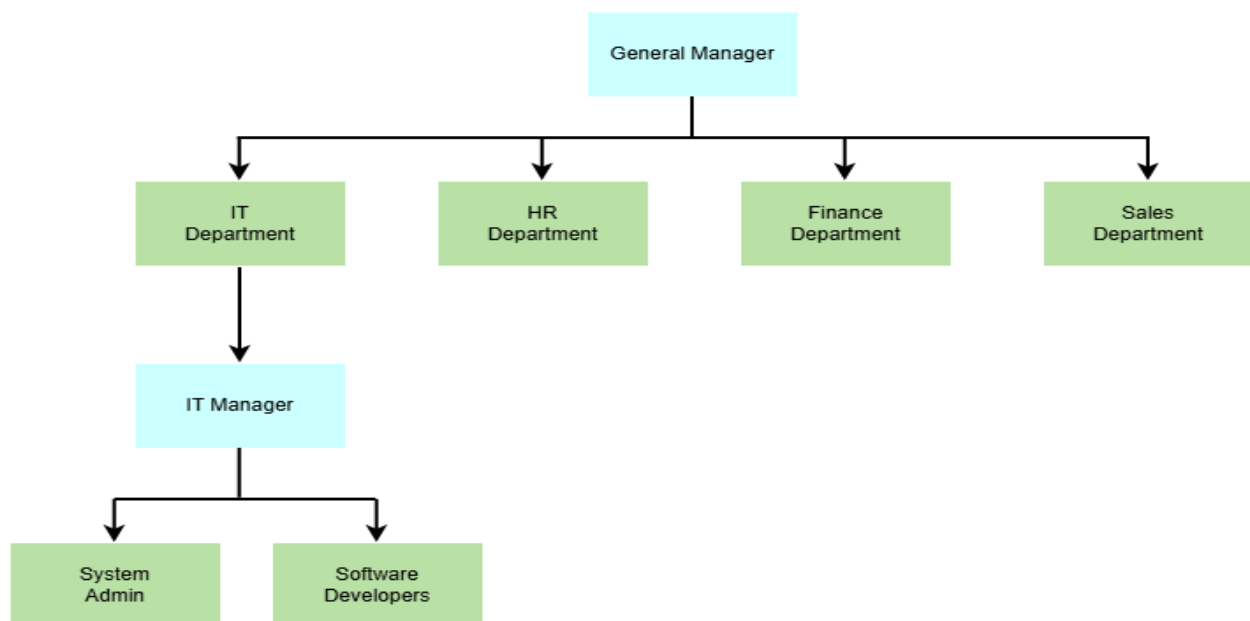
Company Mission:

- ABC Startup Solutions aims to provide high-quality software and IT solutions through skilled employees, efficiency, modern technologies, secure software, and customer first services. The company is committed to improve, innovate, and be a responsible in using technology for its services.

Office Location:

- ABC Towers, Santa Rosa, Laguna

Organizational Structure:



Employee Distribution:

Department	Employees	Percentage
IT Department	5	25
HR Department	4	20
Finance Department	5	25
Sales Department	6	30
Total	20	100%

Part 2 – Enterprise Hardware Inventory

Asset ID	Hardware	Quantity	Department	Purpose
HW-01 to HW-015	Desktop Computer	15	IT, HR, Finance, Sales	Daily office work, and software development/management
HW-16 to HW-020	Laptop	5	IT	Portability
HW-021	Server	1	IT	Centralized service, and hosting
HW-022	Router	1	IT	Internet routing, and internet
HW-023	Switch	1	IT	Connecting wired devices to the LAN
HW-024	Printer	1	Shared	Printing, scanning, and photocopying
HW-025 to HW-026	UPS	2	IT	Power protection
HW-027 to HW-028	Wireless Access Point	2	Shared	Wireless connectivity
HW-029	NAS Storage	1	IT	Centralized file storage
HW-030 to HW-031	External Backup Drive	2	IT	Offline backup
HW-032 to HW-51	Monitors	20	IT, HR, Finance, Sales	Display

Part 3 – Enterprise Software Inventory

Software	Version	License	Purpose
Windows 11 Pro	25H2	Paid	Main operating system for employees' desktop computers and laptops. It's needed since it provides business features

			and security controls suitable for workstations.
Ubuntu Server	26.04 LTS	Free	It serves as the main server operating system for hosting services, apps, databases, and other services. It provides a long-term security and maintenance support.
Microsoft Office	Microsoft 365	Paid	Provides productivity applications needed for office operations, such as Word, Excel, Outlook, and Teams
VS Code	1.132.0	Free	Workstation for software development used by IT personnel and developers for writing, editing, and managing source code.
Git	2.51.1.windows.1	Free	Distributes version control systems used by developers to track changes, manage source code, and collaborate with other developers.
GitHub Desktop	3.6.3	Free	Interface for Git and GitHub. It simplifies the repository management.
VirtualBox	7.2.14	Free	Used to create and manage virtual machines for testing operating systems and applications.
Google Chrome	151.0.7922.77	Free	Web browser used to access the internet and browse websites.
Microsoft Defender	Latest 1.457.103.0	Paid, included in Windows 11 Pro	Provides protection against malware, viruses, spyware, and other security threats on windows computers.
AnyDesk	9.7.14	Free	Used by IT technicians to remotely access other computers to provide technical support.
7-Zip	26.02	Free	Used to compress files to smaller sizes and to extract files. It makes file transferring more efficient, and also saves up storage.

Part 4 – Enterprise Network Inventory

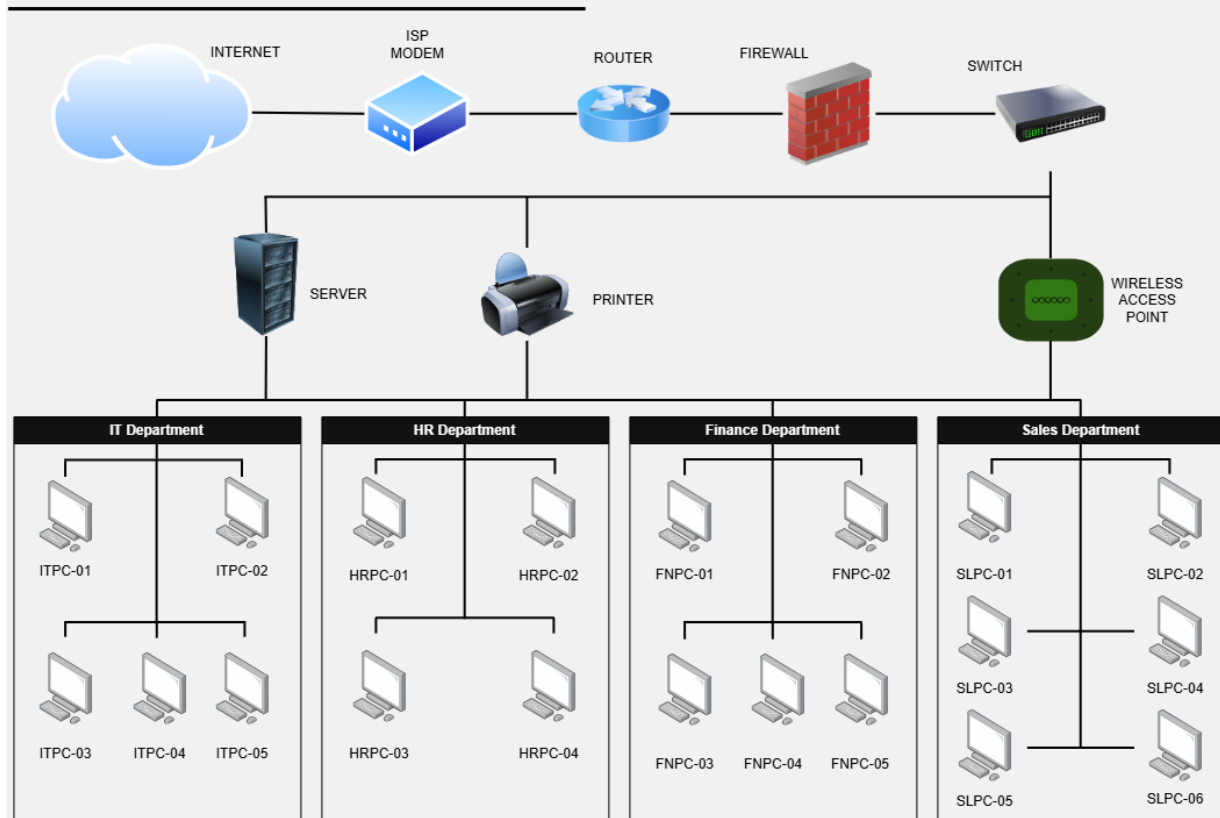
Asset ID	Equipment	Quantity	Purpose
NET-01	ISP Modem	1	This connects the company's network to the ISP's internet service.
NET-02	Router	1	Routes traffic between the local network and internet.
NET-03	Firewall	1	Protects the internal network by filtering unauthorized and harmful network traffic.
NET-04	Switch	1	Connects wired devices like computers, servers, printers within the local network.
NET-05 to NET-06	Access Point	1	Provides wireless network access for wireless devices.
NET-07	Patch Panel	1	Organizes and terminates network cables between the office and the switch.
NET-08 to NET-057	CAT6 Cables	30	Provides reliable wired network connections between the computers and network devices.
NET-058 to NET-107	RJ45 Connectors	50	Used to terminate CAT6 cables and create Ethernet network connections.

Part 5 – Enterprise Network Diagram

ABC STARTUP SOLUTIONS

ENTERPRISE NETWORK TOPOLOGY

MA. LENA THERESE S. QUIZON



Part 6 – System Administration Roles

Helpdesk Technician

1. Responsibilities

- **Technical assistance:** Resolve hardware, software, and network issues reported by users.
- **User engagement:** Communicate with users and address their technical concerns.
- **Problem Identification:** Identify the cause of the problem and determine the best solution.
- **Record Maintenance:** Document user issues, troubleshooting steps, and solutions provided.
- **Training and Onboarding:** Help users learn new systems and technologies.
- **Team Collaboration:** Work with other IT staff to resolve issues and improve support services.

2. Skills

- Knowledge of ticketing systems: Ability to manage, update, and track user support requests.
- Cybersecurity: Knowledge of basic security practices for protecting systems and data.
- Proficiency in software and hardware components: Ability to work with common hardware and software components.
- Troubleshooting: Identify and resolve common technical problems.
- Basic programming: Basic understanding of programming concepts.
- IT diagnostics: Ability to make use of diagnostic tools to identify system and hardware issues.

3. Common Tools

- Ticketing systems
- Remote desktop software
- Active directory
- Command line tools
- VPNs
- Documentation tools

4. Certifications

- Certified Information Systems Security Professional (CISSP)
- IT Infrastructure Library
- Cisco Certified Network Associate
- CompTIA A+
- CompTIA Network +
- VMWare Certification
- Google IT Support Professional Certificate

Network Administrator

1. Responsibilities

- Configure network hardware: Setup and configure routers, switches, firewalls, and other network devices.
- Handling repairs and maintenance: Troubleshooting network problems and performing regular maintenance.
- Designing network models: Planning and designing network structures based on what the organization needs.
- Managing software: Install, configure, and update network-related software and tools.
- Ensuring network security: Protect the network from unauthorized access and security threats.

- Maintaining servers: Monitor, configure, and maintain servers to ensure reliable operation.
- Evaluating system performance: Monitor network performance and resolve issues that affect efficiency of the system.

2. Skills

- Problem-solving: Ability to identify and solve problems effectively.
- Time management: Prioritize tasks and complete them within specific time frames.
- Communication: Effective communication and clearly explaining technical information to users and team members.
- Commitment to learning: Willingness to learn new networking technologies and practices.
- Patience: Ability to remain calm when dealing with complex and recurring technical issues.
- Technical skills: Knowledge in networking concepts, hardware, and protocols, and configurations.
- Teamwork: Ability to efficiently work with other IT staff and departments.
- Attention to detail: Carefully monitoring configurations and identify potential problems.

3. Common Tools

- Network scanning tools: Discover connected devices, open ports, and available services.
- Network protocol analyzers: Capture and examine network traffic to identify problems.
- Process enumeration tools: Identify running and unauthorized processes on networked systems.
- Wireless tools: Monitor, configure, and troubleshoot wireless connections.

4. Certifications

- Cisco Certified Network Associate
- Certified Network Computer Technician
- Cisco Certified Network Professional Enterprise
- Cisco Certified Technician
- CompTIA+
- Microsoft Certified Administrator
- Microsoft Certified Solution Architect
- Microsoft Certified Azure Security Engineer Associate
- Network 5 Certification
- VMWare Certified Professional
- Pro: Windows Server 2008, Enterprise Administrator

Linux System Administrator

1. Responsibilities

- Linux File Systems: Knowledge in understanding different Linux file systems used by Linux. Manage Linux storage, partitions, and file permissions.
- File System Hierarchy: Knows how to keep files and directories properly organized.
- Managing Root/Super User: Control administrative access to protect important system files and maintain system security.
- Basic Bash Command: Linux System Administrators should have at least a basic knowledge of how to use bash commands to facilitate system management and administrative tasks efficiently.
- Handling File, Directories, and Users: Should have basic knowledge on how to create, edit, organize, and manage basic file and directory management tasks.

2. Skills

- User account management: Create and manage user accounts, groups, and passwords while ensuring specific users only has access to resources they need.
- SQL: Knowledge in SQL to retrieve, organize, manage database information, as well as identifying basic database issues.
- Backup and restore: It is an essential skill to be able to backup, restore, and be able to do disaster recovery for a Linux System Administrator.
- Hardware setup and troubleshooting: Linux SAs need to have knowledge in setting up, maintaining, and troubleshooting hardware systems to keep systems running properly.

3. Common Tools

- Monitoring and performance tools: Linux System Administrators use these tools to track system performance and resource to identify issues and keeps systems running efficiently.
- User and permission management tools: Control access to system resources by managing user accounts, groups, passwords, and permissions.
- Networking tools: These tools are used to configure, monitor, and troubleshoot network connections, interfaces, and services.
- Security and firewall tools: Assist in protecting linux systems by controlling network traffic and preventing unauthorized access.
- Backup and recovery tools: Administrators rely on these tools to create, manage, and restore backups to protect important data from loss of system failures.

4. Certifications

- CompTIA Linux+
- Red Hat Certified System Administrator (RHCSA)
- Linux Foundation Certified Engineer (LFCE)
- Red Hat Certified Engineer (RHCE)
- AWS Certified SysOps Administrator
- Kubernetes and Cloud Native Certifications (CKAD, CKA)

Cloud Administrator

1. Responsibilities

- Manage cloud systems and infrastructure: Cloud administrators configure, maintain, and monitor cloud resources, applications, and services to ensure reliable performance and security.
- Design internal data policies and processes: It's usually the cloud administrator's duty to develop policies and procedures for managing cloud data, resources, and security within the organization.
- Conduct training sessions for colleagues: They train and guide employees in using cloud services properly and following security practices.
- Oversee cloud network and security: Cloud administrators monitor cloud networks and security controls to identify issues and protect the company's resources.
- Troubleshoot operational and performance issues: They are usually responsible for monitoring, identifying, and resolving cloud-related problems involving access, performance, updates, and system operations.

2. Skills

- Technical expertise: Being a cloud administrator requires the understanding of cloud technologies such as networking, virtualization, automation, storage, operating systems, and data management.
- Problem-solving: Should be able to analyze and resolve issues related to cloud installation, deployment, migration, maintenance, and performance.
- Communication: Clearly communicate and collaborate with developers, engineers, and other members since they usually work together.
- Time management: It is important for cloud administrators to learn and adjust to new cloud technologies, services, tools, and changes in the organization's requirements. This means, they are expected to handle multiple tasks effectively while adapting to new technologies.

3. Common Tools

- **Cloud Platforms:** Cloud platforms like AWS, Azure, and Google Cloud Platform provides infrastructure and services needed to host, manage, and run applications and resources in the cloud.
- **Infrastructure as Code (IaC):** These platforms help automate the deployment and manage of cloud infrastructures using configuration files and codes.
- **Cloud Security and Compliance Tools:** These tools protect cloud resources by managing access, monitor security risks, and maintaining standards in security.
- **Monitoring and Performance Management Tools:** Administrators use monitoring and performance management tools to track cloud resources, system performance, and application activity to maintain peak cloud performance levels.
- **Backup and Disaster Recovery Tools:** Backup and disaster recovery tools protect important cloud data and services by creating backups and ensuring there are recovery options during system failures.

4. Certifications

- AWS Certified Developer – Associate
- Microsoft Certified: Azure Administrator
- AWS Certified Cloud Practitioner
- Google Cloud Associate Cloud Engineer
- Professional Cloud Architect (Google)

These four works together in an organization to keep the organization's IT infrastructure reliable, efficient, effective, and secure. The Helpdesk Technician acts as the frontline support to company employees; they are tasked with handling the common issues faced by the employees. If they themselves faced a complex technical problem, then they will refer it to an appropriate IT professional. The Network Administrator handles all the networking components or issues being faced by the company. Meanwhile, the Linux System Administrator manages Linux-based servers, accounts, system services, and data backups. Lastly, the Cloud Administrator handles all cloud related services. They may have different specializations, but once they work together, they form a coordinated team that supports the organization's entire technological infrastructure.

Part 7 – Infrastructure Recommendations

1. Internet Provider

- ABC Startup Solutions should subscribe to a dependable fiber broadband with at least a speed of 300 Mbps. A business-grade connection is highly

recommended since the company relies on cloud services, development tools, file transfers, and other online transactions. The connectivity to fiber broadband provides better stability, and lower latency than many alternative connections.

2. Server Specifications

- It's recommended that a company like ABC Startup Solutions should use a dedicated server with a greater number of CPU cores, recommended brands are AMD EPYC and Intel Xeon. As for the RAM, higher RAM is best for large-scale databases and enterprises environments. A storage capacity more than 2T is highly recommended for tech companies as that much storage is needed for databases and media-heavy applications.

3. Backup Strategy

- The company should apply a 3-2-1 backup rule. Three is for "three copies of data", wherein you have the original copy, and then a backup of two copies. Two is for "two storage types", if one device stops working or gets damaged, the company still has another copy of data available. Lastly, one is for "one off-site copy". Keeping a copy in another location like a cloud storage, it protects the data if something happens within the office.

4. Security Recommendations

- The company should implement a firewall, secure Wi-Fi configuration, regular software update, access controls, and network segmentation. Employee accounts should also use appropriate permissions based on their roles within the company. Important infrastructure and systems should only be accessed by the higher ups and the authorized IT personnel

5. Antivirus

- The Microsoft defender antivirus should be used in partner with the windows 11 Pro computers that is planned for the company. It provides a built-in protection against malware, viruses, spyware, and other cyber threats.

6. Password Policy

- Companies should require each employee to use strong passwords with at least 8 to 12 characters long with not only letters, but also with numbers and special characters. They should avoid passwords related to their names and birthdays as those can be easily guessed by others.

7. Expansion Plan

- The company should also invest and plan for future growth by choosing hardware, software, and network components that can be easily upgraded and expanded once it is needed. Planning ahead will make it easier and

less expensive to add new employees, equipment, and services without replacing the entire system.

Part 8 – Personal Reflection

The making of this helped me get a better understanding on how important the role of a system administrator is. I learned how important it is to properly plan an organization's IT infrastructure based on what the company's needs. Planning the whole hardware, software, and the network inventories gave me a better understanding how devices and technologies play a key role in an organization.

The most challenging part was creating the inventory since I need to be able to choose realistic quantities and the right equipment for a company with just 20 employees. It was also a bit of a challenge to design the network topology. The different devices and network components should be carefully thought of, and how they all should be connected to each other, and how each department would access the network. This made me realize that system administrators need to consider both the current and possible future requirement of an organization.

Before the deployment, it is important that a proper planning is held since it will help prevent mistakes, unnecessary expenses, and problems after the system is installed. A proper plan allows an organization to determine what hardware, software, network equipment, security measures, and backup solutions are needed before purchasing or configuring anything. A plan will also make it easier to identify possible problems and prepare solutions in advance. If there is no proper planning, the company could possibly make mistakes in purchasing equipment that does not meet its needs or even build a network that will be difficult to maintain and expand in the future.

Overall, this project helped me develop a more organized and practical approach to handling IT-related tasks. It has taught me that being a System Administrator also requires a person to have good decision-making skills, attention to details, and the ability to prepare for possible problems in the future. The knowledge and experience I have gained from making this project can help me become more confident when working with actual systems and handling the responsibilities that a system administrator has with managing an organization's IT infrastructure.